Gyumin Kim

M.S.–Ph.D. Student, Semiconductor Design Automation (SeDA) Lab, UNIST

q.min@unist.ac.kr · gyum-lab.github.io · Google Scholar · LinkedIn

Research Interests

- **Physical design methodologies for 3D integration** — hybrid-bonded F2F 3D ICs (tier partitioning, 3D via legalization), unified flows for homogeneous and heterogeneous 3D ICs, monolithic 3D ICs (thermal-aware placement, cross-tier routability)
- **Backside design methodologies for advanced technology nodes** — backside power delivery network (BS-PDN) optimization for F2F 3D ICs, backside clock distribution network (BS-CDN) synthesis
- **Machine learning for EDA (ML-EDA)** — reinforcement learning for routing congestion mitigation and 3D via legalization, graph neural networks for physical design, learning-based prediction for placement refinement and thermal analysis

Education

Ulsan National Institute of Science and Technology (UNIST) Sep 2024 – Present
M.S.–Ph.D. Integrated Program in Electrical Engineering
Semiconductor Design Automation (SeDA) Lab, Advisor: Prof. Heechun Park

Kookmin University Aug 2024
B.S. in Electrical Engineering

Research Experience

SeDA Lab, UNIST, Ulsan, Korea Sep 2024 – Present
Graduate Researcher (Advisor: Prof. Heechun Park)

Dept. of ECE, University of Southern California, Los Angeles, CA Sep 2025 – Oct 2025
Visiting Researcher (Host: Prof. Sung Kyu Lim), two-week research visit (Sep 22 – Oct 4, 2025)

SeDA Lab, UNIST, Ulsan, Korea Mar 2024 – Aug 2024
Visiting Researcher (Advisor: Prof. Heechun Park)

SeDA Lab, Kookmin University, Seoul, Korea Sep 2023 – Feb 2024
Undergraduate Researcher (Advisor: Prof. Heechun Park)

Publications

Journal Articles

[J1] H. Jung, J. Choi, S. Baek, B. G. Shin, Y. J. Song*, H. Jung, J. Kim, J. Jeon*, **Gyumin Kim**, Heechun Park*, Y. Lee, J. Yoo*, J.-H. Lee, H. Kim, K. Kang*, J. Jeong, S. H. Kim*, J. Bae, C. S. Kim, W. K. Yang, S. Lee, J. Kwon*, B.-S. Kim, J.-H. Han*, H.-J. Kim, H. H. Yoon*, J. Kwon*, and Y. J. Hong*, "Advances and Future Challenges in Monolithic 3D Integrated Logic, Power, and Optoelectronics Technologies for Tightly Interconnected Intelligent Systems," *ACS Nano*, vol. 20, no. 8, pp. 6407–6445, 2026.

Conference Papers

[C1] **Gyumin Kim** and Heechun Park, "Power Network Topology Optimization for Top Dies in Backside-Powered F2F 3D ICs," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2027. (Accepted)

[C2] Ikkyum Kim, Junghyun Yoon, **Gyumin Kim**, Jooyeon Jeong, and Heechun Park, "Comprehensive BS-CDN Synthesis Framework with GNN Clustering and MILP-Based Optimization," *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026. (Accepted)

[C3] **Gyumin Kim** and Heechun Park, "Differentiable Thermal-Aware Placement for Monolithic 3D ICs with Stack-Aware Thermal Modeling," *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026. (Accepted)

[C4] **Gyumin Kim** and Heechun Park, "Prediction-Guided Placement Refinement for Cross-Tier Routability in Monolithic 3D ICs," *ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)*, 2026.

[C5] **Gyumin Kim** and Heechun Park, "Segment-Level Rerouting with Reinforcement Learning for Routing Congestion Mitigation," *ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)*, 2026. **(Best Paper Award)**

[C6] Junghyun Yoon, Ikkyum Kim, **Gyumin Kim**, Sojung Park, and Heechun Park, "Standard Cell Layout Generation: Methodological Evolution and Architectural Impacts," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026.

[C7] **Gyumin Kim** and Heechun Park, "A Unified Design Flow for Homogeneous and Heterogeneous 3D Integration with Fine-Pitch Hybrid Bonding," *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2025.

[C8] **Gyumin Kim** and Heechun Park, "Clustering-Driven Bonding Terminal Legalization with Reinforcement Learning for F2F 3D ICs," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025.

Teaching

Digital Logic and Laboratory (EEE20201, EEE20202), UNIST Fall 2024
Teaching Assistant

Honors & Awards

Best Paper Award, MLCAD 2026 Sep 2026
ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)
"Segment-Level Rerouting with Reinforcement Learning for Routing Congestion Mitigation"

Skills

EDA Tools	Cadence Innovus, Genus, Celsius; Synopsys ICC2, Design Compiler, PrimeTime, VCS, Verdi
Programming	Python, C/C++, Tcl, PyTorch
Tools & OS	Linux